Nicola Bellomo

Curriculum Vitae

Dipartimento di Fisica "Galileo Galilei",
Universitá degli Studi di Padova,
Padova, 35131, Italy
✉ *nicola.bellomo@unipd.it*
ORCID ID: 0000-0002-4375-705X

Personal Information

Full Name **Nicola Bellomo**.
Place of Birth **Feltre (Italy)**.
Date of Birth **11/06/1991**.
Nationality **Italian**.

Research Interests

Large scale structure at low and high redshift - Gravitational waves - Dark matter modelling - Primordial black holes - Early universe cosmology.

Current position

Oct 2023 - now **Postdoctoral Fellow**, *Dipartimento di Fisica "Galileo Galilei", Universitá degli Studi di Padova*, Padova, Italy.

Previous positions

Oct 2020 - Sep 2023 **Postdoctoral Fellow**, *Texas Center for Cosmology and Astroparticle Physics, Weinberg Institute, The University of Texas at Austin*, Austin, USA.

Feb 2020 - Sep 2020 **Postdoctoral Fellow**, *ICCUB - University of Barcelona*, Barcelona, Spain.

Education

Jan 2016 - Jan 2020 **Ph.D. Degree in Physics**, *ICCUB - University of Barcelona*, Barcelona.
Thesis title: *Exploring Signatures of New Physics in Cosmology*. Supervisor: Prof. Licia Verde. Evaluated as "Excellent cum laude".

Oct 2013 - Oct 2015 **Master Degree in Physics**, *University of Padova*, Padova, 110/110.
Thesis title: *Application of the Excursion Set Theory to Modified Gravity*. Supervisor: Prof. Sabino Matarrese, Dr. Philippe Brax.

Oct 2010 - Sep 2013 **Bachelor Degree in Physics**, *University of Padova*, Padova, 107/110.
Thesis title: *Superfluid Hydrodynamics*. Supervisor: Prof. Luca Salasnich.

Honours and awards

2025-2027 **Marie Skłodowska-Curie Action - European Fellowship**, *Project: GWSKY - Mapping the Universe with Gravitational Waves, 210k€.*

2025 **Beatriu de Pinos Postdoctoral Fellowship**, *161k€ - Declined.*

Publications

Submitted and Online Papers

2609.09083 **[2] N. Bellomo, M. Bosi, S. Libanore, M. Liguori, M. Mapelli, F. Semenzato, S. Torniamenti**, *Breaking binary formation mechanism degeneracies with gravitational wave clustering*.

2607.12049 **[1] C. Scarpel, N. Bellomo, S. Libanore, M. Liguori, A. Raccanelli**, *Mitigating galaxy systematics with gravitational wave clustering*.

Refereed Papers

2510.24857 **[22] F. Semenzato, N. Bellomo, C. M. F. Mingarelli, A. Raccanelli**, *Bias from small-scale leakage in Pulsar Timing Array maps*, JCAP **09** (2026), 021.

2512.14668 **[21] G. Perna, N. Bellomo, V. Roatti, D. Bertacca**, *From the Solar System to cosmological distances: a complete formalism for gravitational wave astrometry*, JCAP **07** (2026), 102.

2605.08080 **[20] G. Montefalcone, N. Bellomo, K. K. Boddy**, *CMB Limits on the Absorption of Light Vector and Axial-Vector Dark Matter*, PRD **114**, 023548 (2026).

2411.00532 **[19] F. Semenzato, J. Andrew Casey-Clyde, C. M. F. Mingarelli, A. Raccanelli, N. Bellomo, N. Bartolo, D. Bertacca**, *Cross-Correlating the Universe: The Gravitational Wave Background and Large-Scale Structure*, JCAP **06** (2026), 066.

2502.08562 **[18] E. Vanetti, E. Vanzan, N. Bellomo, A. Raccanelli**, *Testing the growth of cosmic structures during the Dark Ages*, JCAP **09** (2025), 054.

2503.00110 **[17] G. Montefalcone, G. Elor, K. K. Boddy, N. Bellomo**, *CMB Constraints on Loop-Induced Decays of Leptophilic Dark Matter*, PRD **112**, 023506 (2025).

2405.13871 **[16] E. Vanzan, S. Libanore, L. Valbusa Dall'Armi, N. Bellomo, A. Raccanelli**, *Gravitational wave background from primordial black holes in globular clusters*, JCAP **10** (2024), 014.

2306.03031 **[15] M. Bosi, N. Bellomo, A. Raccanelli**, *Constraining extended cosmologies with GW×LSS cross-correlations*, JCAP **11** (2023), 086.

2210.15691 **[14] N. Bellomo, K. V. Berghaus, K. K. Boddy**, *Impact of freeze-in on dark matter isocurvature*, JCAP **11** (2023), 024.

2210.14934 **[13] L. Piga, M. Lucca, N. Bellomo, V. Bosch-Ramon, S. Matarrese, A. Raccanelli, L. Verde**, *The effect of outflows on CMB bounds from Primordial Black Hole accretion*, JCAP **12** (2022), 016.

2110.15059 **[12] N. Bellomo, D. Bertacca, A. C. Jenkins, S. Matarrese, A. Raccanelli, T. Regimbau, A. Ricciardone, M. Sakellariadou**, *`CLASS_GWB`: robust modeling of the astrophysical gravitational wave background anisotropies*, JCAP **06** (2022), 030.

2005.09666 **[11] J. L. Bernal, N. Bellomo, A. Raccanelli, L. Verde**, *Beware of commonly used approximations II: estimating systematic biases in the best-fit parameters*, JCAP **10** (2020), 017.

2005.10384 **[10] N. Bellomo, J. L. Bernal, G. Scelfo, A. Raccanelli, L. Verde**, *Beware of commonly used approximations I: errors in forecasts*, JCAP **10** (2020), 016.

2004.11224 **[9] V. Bosch-Ramon, N. Bellomo**, *Mechanical feedback effects on primordial black hole accretion*, A&A **638** A132 (2020).

1907.03660 **[8] A. R. Khalifeh, N. Bellomo, J. L. Bernal, R. Jimenez**, *Can Dark Matter be Geometry? A Case Study with Mimetic Dark Matter*, PDU **30** 100646 (2020).

1909.11627 **[7] D. Bertacca, A. Ricciardone, N. Bellomo, A. C. Jenkins, S. Matarrese, A. Raccanelli, T. Regimbau, M. Sakellariadou**, *Projection effects on the observed angular spectrum of the astrophysical stochastic gravitational wave background*, PRD **101** 103513 (2020).

1908.03596 **[6] A. Kalaja, N. Bellomo, N. Bartolo, D. Bertacca, S. Matarrese, I. Musco, A. Raccanelli, L. Verde**, *From Primordial Black Holes Abundance to Primordial Curvature Power Spectrum (and back)*, JCAP **10** (2019), 031.

1809.07113 **[5] N. Bellomo, N. Bartolo, R. Jimenez, S. Matarrese, L. Verde**, *Measuring the Energy Scale of Inflation with Large Scale Structures*, JCAP **11** (2018), 043.

1809.03528 **[4] G. Scelfo, N. Bellomo, A. Raccanelli, S. Matarrese, L. Verde**, *GW×LSS: Chasing the Progenitors of Merging Binary Black Holes*, JCAP **09** (2018), 039.

1709.07467 **[3] N. Bellomo, J. L. Bernal, A. Raccanelli, L. Verde**, *Primordial Black Holes as Dark Matter: Converting Constraints from Monochromatic to Extended Mass Distributions*, JCAP **01** (2018), 004.

1709.07465 **[2] J. L. Bernal, N. Bellomo, A. Raccanelli, L. Verde**, *Cosmological implications of Primordial Black Holes*, JCAP **10** (2017), 052.

1612.02598 **[1] N. Bellomo, E. Bellini, B. Hu, R. Jimenez, C. Peña-Garay, L. Verde**, *Hiding neutrino mass in modified gravity cosmologies*, JCAP **02** (2017), 043.

Books

2606.20366 **[2] Bonaldi et al., including N. Bellomo**, *Advancing Astrophysics with the SKA II*, Member of the editor team.

2312.14097 **[1] V. De Luca, N. Bellomo**, *The accretion, emission, mass and spin evolution of primordial black holes*, Chapter of the book "Primordial Black Holes", ed. Chris Byrnes, Gabriele Franciolini, Tomohiro Harada, Paolo Pani, Misao Sasaki; Springer (2025).

White Papers

2606.25671 **[7] N. Bellomo, P. Cole**, *Exploring Gravitational Wave Science Frontiers with the SKAO*, Chapter of the "Advancing Astrophysics with the SKA II" book.

2606.24992 **[6] G. Perna, N. Bellomo, D. Bertacca**, *Synergies between pulsar timing array and astrometry*, Chapter of the "Advancing Astrophysics with the SKA II" book.

2606.24846 **[5] M. Bosi, S. Libanore, N. Bellomo, C. Scarpel, F. Semenzato, A. Raccanelli, M. Liguori**, *Using SKAO to understand the clustering of gravitational wave sources*, Chapter of the "Advancing Astrophysics with the SKA II" book.

2503.12263 **[4] A. Abac et al.**, *The Science of the Einstein Telescope*, 2025.

2204.05434 **[3] E. Auclair et al.**, *Cosmology with the Laser Interferometer Space Antenna*, 2022.

2001.09793 **[2] E. Barausse et al.**, *Prospects for Fundamental Physics with LISA*, 2020.

1903.04424 **[1] Y. Ali-Haïmoud et al.**, *Electromagnetic probes of primordial black holes as dark matter*, White Paper submitted to the Astro-2020 (2020 Astronomy and Astrophysics Decadal Survey), 2019.

Proceedings

[1] N. Bellomo, G. Scelfo, A. Raccanelli, S. Matarrese, L. Verde, *The origin of progenitors in merging black hole binaries*, Highlights on Spanish Astrophysics X (2019).

Teaching Experience

Spring 2026 **Teaching Assistant for "Physics with Applications to Biological Systems"**, *University of Padova*, Bachelor Degree in Biology of Human and Environmental Health.

Fall 2025 **Teaching Assistant for "Mathematical and Numerical Methods"**, *University of Padova*, Master Degree in Astrophysics and Cosmology.

Spring 2025 **"Introduction to Gravitational Waves" - 4 hours mini-course**, *University of Padova*, Bachelor Degree in Physics.

Spring 2018 - Spring 2020 **Teaching assistant for "Differential Equations and Vector Calculus"**, *University of Barcelona*, Bachelor Degree in Physics.

Spring 2017 **Teaching assistant for "Differential Equations and Vector Calculus"**, *University of Barcelona*, Bachelor Degree in Electronic Engineering.

Fall 2016 - Fall 2019 **Teaching assistant for "Theoretical Mechanics"**, *University of Barcelona*, Bachelor Degree in Physics.

Mentoring Experience

PhD Students

2023-2026 **Co-advisor of Jessie de Kruijf**, *University of Padova*.

2023-2026 **Co-advisor of Federico Semenzato**, *University of Padova*, later Postdoc at Johns Hopkins.

2023-2025 **Co-advisor of Francesco Spezzati**, *University of Padova*, later Postdoc at Caltech.

Master Students

2024-now **Co-advisor of Granite Shkodra**, *University of Padova*.

2024-2025 **Co-advisor of Myrto Magklara**, *University of Padova*, now PhD at University of Ioannina with Leandros Perivolaropoulos.

2024 **Co-advisor of Francesco Pozzi**, *University of Padova*.

2023-2024 **Co-advisor of Caterina Scarpel**, *University of Padova*.

2023-2024 **Co-advisor of Elena Vanetti**, *University of Padova*, now PhD student at IAC with Carlos Hernandez Monteagudo.

2021-2022 **Co-advisor of Michele Bosi**, *University of Padova*, now PhD student at SISSA with Carlo Baccigalupi.

2019-2020 **Co-advisor of Lorenzo Piga**, *Institute of Cosmos Sciences*, later PhD student at the University of Parma with Guido D'Amico.

2018-2019 **Co-advisor of David Figueruelo**, *Institute of Cosmos Sciences*, later PhD student at University of Salamanca with José Beltrán Jiménez.

2018-2019 **Co-advisor of Alba Kalaja**, *Institute of Cosmos Sciences*, later PhD student at University of Groeningen with Daan Meerburg.

2017-2018 **Co-advisor of Giulio Scelfo**, *Institute of Cosmos Sciences*, later PhD student at SISSA with Matteo Viel.

Undergraduate Students

2026-now **Co-advisor of Benedetta Cagnoni**, *University of Padova*.

2025-now **Co-advisor of Maria Pizzamano**, *University of Padova*.

2025-2026 **Co-advisor of Filippo Chinchio**, *University of Padova*.

2024 **Co-advisor of Damiano Cerioni**, *University of Padova*.

Community Service

Referee for International Journals

since 2023 **Physical Review D**.

since 2022 **Physical Review Letters**.

since 2021 **The Astrophysical Journal, Monthly Notices of the Royal Astronomical Society**.

since 2019 **Modern Physics Letters A**.

since 2018 **Journal of Cosmology and Astroparticle Physics, Europhysics Letters**.

since 2017 **Physics of the Dark Universe**.

Organizational Duties

2021-2023 **Organizer of the Weinberg Institute Seminars**.

2018-2020 **Founder and organizer of the ICC "Coffe, PhD & Science Meetings"**.

2016-2020 **PhD Students representative in ICC Board**.

Major Research Collaborations

since 2019 **Member of SKA Gravitational Waves Working Group**.

since 2024 **Co-Chair of SKA Gravitational Waves Working Group**.

since 2024 **Member of Einstein Telescope Cosmological Division**.

Workshop Organization

24-28 Nov '25 **SKA-Italy Workshop**, *University of Bologna*, Scientific Organizing Committee.

16-20 Jun '25 **SKAO General Science Meeting 2025**, *German Center for Astrophysics - DZA*, Scientific Organizing Committee.

14-16 Apr '25 **UniVersum - VI Edition**, *University of Padova*, Local Organizing Committee.

23-26 Sep '24 **COSMO CLASSIC 2024: future prospects for Cosmology**, *Osservatorio Astronomico di Asiago*, Local Organizing Committee.

20-21 Jan '20 **Barcelona-Valencia Meeting on Gravitational Waves and Virgo**, *Institute of Cosmos Sciences - University of Barcelona*, Local Organizing Committee.

04-07 Jun '18 **Venice Cosmology Workshop 2018: The Island**, *Venice International University*, Local Organizing Committee.

Conferences and Seminar

Scientific Contributions

9 Sep '26 **Breaking degeneracies in BBH formation mechanisms**, *Listening to the Cosmos*, Firenze.

17 Jun '26 **Breaking degeneracies in BBH formation mechanisms**, *1st BiCoq Conference*, Milano.

8 May '26 **Breaking degeneracies in BBH formation mechanisms**, *Universum VII*, Napoli.

23 Apr '26 **Bias from small-scale leakage in Pulsar Timing Array maps**, *GRaviCon 2026*, Scuola Normale Superiore.

19 Jun '25 **Pulsar and gravitational waves early science with SKAO**, *SKAO Science Meeting 2025 (invited panellist)*, Görlitz.

23 May '25 **Gravitational wave science in the SKAO era**, *Majorana-Raychaudhuri Seminar (invited)*, Online.

10 Mar '25 **Gravitational wave science in the SKAO era**, *Seminar*, ICCUB, Barcelona.

6 Mar '25 **Gravitational wave science in the SKAO era**, *Severo Ochoa seminar (invited)*, IAA, Granada.

5 Mar '25 **Gravitational wave science in the SKAO era**, *Seminar*, IFT, Madrid.

23 Oct '24 **Primordial Black Holes in globular clusters**, *GraSP24 (invited talk)*, University of Pisa, Pisa.

7 Feb '24 **Probing dark matter - baryon scattering using cosmological observations**, *Cosmology Journal Club*, University of Padova, Padova.

14 Nov '22 **Understanding the nature of dark matter**, *Hu-Joyce group meeting*, University of Chicago, Chicago.

4 Nov '22 **Understanding the nature of dark matter**, *Astro seminar (invited)*, University of South California, Los Angeles.

10 Oct '22 **A different perspective on Primordial Black Holes**, *Theoretical Astroparticle and Cosmology Symposium in Texas (invited)*, Southern Methodist University, Dallas.

24 Jun '21 **Tracing the large scale structure of the Universe using gravitational waves**, *CosKASI ECR seminar (invited)*, KASI, Daejeon.

27 Oct '20 **Searching for Primordial Black Holes**, *Theory Group Seminars*, UT-Austin, Austin.

20 Jan '20 **GWxLSS: Combining large scale structure and gravitational waves**, *Barcelona-Valencia Meeting on Gravitational Waves and Virgo*, ICCUB, Barcelona.

11 Jun '19 **The nature of progenitors of merging binary black holes**, *España en SKA!*, IAA, Granada.

08 May '19 **What can we learn from Primordial Black Holes?**, *Cosmo Coffee*, CERN, Geneve.

20 Mar '19 **The nature of progenitors of merging binary black holes**, *Internal Seminars*, SISSA, Trieste.

10 Oct '18 **The nature of progenitors of merging binary black holes**, *Internal Seminars*, ICCUB, Barcelona.

17 Jul '18 **The origin of progenitors in merging black hole binaries**, *XIII Reunión Científica de la SEA*, University of Salamanca, Salamanca.

05 Jul '18 **Non-Gaussianity and Large Scale Structures**, *Unsolved Problems in Astrophysics and Cosmology (invited)*, Eötvös University, Budapest.

06 Jun '18 **Axion Dark Matter**, *Venice Cosmology Workshop 2018: The Island*, Venice International University, Venice.

01 Feb '18 **Primordial Black Holes in Cosmology**, *Winter Meeting (invited)*, ICCUB, Barcelona.

12 Sep '17 **Hiding neutrino mass in modified gravity cosmologies**, *Meeting on Fundamental Cosmology*, CEFCA, Teruel.

Professional Contributions

15 Jul '20 **Coffee get-together on professional prospects**, *XVI Reunión Científica (virtual) de la SEA (invited)*.

Outreach

Sep 2025 **Science4All**, *Palazzo Bo*, Member of the Department of Physics and Astronomy team.

Apr 2025 - Jun 2025 **Le scale dell'Universo**, *Department of Physics and Astronomy "Galileo Galilei"*, Member of the installation design team.

2024-2026 **Premio Asimov**, Member of the Veneto Scientific Committee.

Sep 2020 **Introducción a la Cosmología**, *Centre Cívic del Guinardó*, Outreach event for the association "Sopa de particulas".

Oct 2019 **Matèria Fosca**, *University of Barcelona*, Round table for undergraduate students on the occasion of the 2019 "Dark Matter Day".

2019 - 2020 **The End of the Universe**, *University of Barcelona*, Artistic project by Andy Gracie about astrophysics and cosmology.

May 2019 **Viatge cap a l'Univers fosc**, *Museu de Sant Cugat*, Presentation of the exposition "Unravelling the dark universe".

Jan-Dec 2019 **Treball de Recerca**, *University of Barcelona*, Advising two high school students during the realization of their final project ("treball de recerca").

Jan 2019 **Experiments de Física**, *University of Barcelona*, Activity for fifth year high school students interested in Astronomy and Astrophysics.

Oct 2018 **Matèria Fosca**, *University of Barcelona*, Round table for undergraduate students on the occasion of the 2018 "Dark Matter Day".

Jun 2018 **CampusPhys 2018**, *University of Barcelona*, Activity for fourth year high school students interested in Astronomy and Astrophysics.

Jan 2018 **Experiments de Física**, *University of Barcelona*, Activity for fifth year high school students interested in Astronomy and Astrophysics.

Jan-Jun 2017 **Unravelling the dark universe**, Exposition for high school students and general public about dark matter, dark energy, black holes and gravitational waves.

Nov 2016 **Encontres amb el tercer cicle: Merging into the darkness**, *University of Barcelona*, Outreach event for undergraduate students of the Physics department.

Press Release

Mar 2025 **Videointerview on SKAO impact on Gravitational Wave Science**, *IAA - Granada*, Available at link.